

PROBLEM SET 5

ECO403/503–INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

Instructions: Due on 28/11/19 (optional) before the final starts. This problem set is optional. I will take the average over four problem sets to compute your assignments grade. If you deliver this one it will be extra. You are encouraged to work in groups but each student is required to submit her or his own work. If you worked with a group, please indicate the names of each member in your submission. Don't forget to attach the cover sheet.

1. Consider a principal-agent problem with three exogenous states of nature, θ_1, θ_2 and θ_3 ; two effort levels a_L and a_H ; and two output levels, distributed as follows as a function of the state of nature and the effort level:

State of nature	θ_1	θ_2	θ_3
Probability	0.25	0.5	0.25
Output under a_H	18	18	1
Output under a_L	18	1	1

The principal is risk neutral, while the agent has utility function $\sqrt{w}$ when receiving monetary compensation w , minus the cost of effort, which is normalized to 0 for a_L and to 0.1 for a_H . The agent's reservation expected utility is 0.1.

- (a) Find the first-best contract.
 - (b) Derive the second-best contact when only output levels are observable.
 - (c) Assume the principal can buy for a price of 0.1, an information system that allows the parties to verify whether state of nature θ_3 has happened or not. Will the principal buy this information system?
 - (d) Consider the situation described in the last part about the information system. Suppose now that the price of the information system described above is p . What is the maximum price that the principal would be willing to pay for this information system?
2. Consider the following problem principal-agent problem. The principal is risk neutral. There are three possible actions $E = \{e_1, e_2, e_3\}$. There are two possible profit outcomes: $\pi_H = 10$ and $\pi_L = 0$. The probabilities of π_H conditional on the three effort levels are $f(\pi_H|e_1) = 2/3$, $f(\pi_H|e_2) = 1/2$, and $f(\pi_H|e_3) = 1/3$. The agent's effort cost function has $g(e_1) = 5/3$, $g(e_2) = 8/5$, and $g(e_3) = 4/3$. Finally, the agent's utility over wealth levels is $u(w) = \sqrt{w}$, and the agent's reservation utility is 0.
 - (a) What is the optimal contract when effort is observable?
 - (b) Show that if effort is not observable, then e_2 is not implementable by the principal as an optimal effort level. For what levels of $g(e_2)$ would e_2 be implementable? *Hint: try solving the problem in terms of utility levels.*
 - (c) What is the optimal contract when effort is not observable?
 3. Take a standard moral hazard problem in which there are:
 - two actions $a \in \{0, 1\}$ (e.g., low effort, high effort)
 - two outcomes $q \in \{0, 1\}$ (failure or success)
 - The probability of a success is

- $\Pr(q = 1|a = 0) = p_0 > 0$
- $\Pr(q = 1|a = 1) = p_1 > p_0$
- Agent's payoff if he takes action a and gets paid is w :

$$u(w) - c(a)$$

where $u(0) = 0, u' > 0, u'' \leq 0$ and $c(0) = 0 < \psi = c(1)$

- But now the principal is *risk averse*, i.e. its payoff is

$$v(S(q) - w),$$

where $S(1) = s_1 > s_0 = S(0)$ and $v(0) = 0, v' > 0, v'' < 0$.

- (a) Consider first the case in which the agent is risk neutral, i.e., assume $u(w) = w$.
 - i. Derive the first-best contract.
 - ii. Derive the second-best contract.
- (b) Now consider the case in which the agent is strictly risk averse, i.e., assume $u''(w) < 0$.
 - i. Derive the first-best contract.
 - ii. Derive the second-best contract.
- (c) Discuss the difference that you see compared to the case that we analyzed in the lectures, i.e., the case of a risk neutral principal.